3 Complex Analysis 2024
Tutorial 10

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Real integrals

Problem 41 Evaluate the following real integrals using methods of complex analysis:

(a) $\displaystyle\int_0^{2\pi} \frac{1}{1-2r\cos\theta + r^2}\,d\theta$, where $0 \le r < 1$,

(b) $\displaystyle\int_0^{\infty} \frac{1}{x^4+6x^2+8}\,dx$,

(c) P.V. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{(2-x)(x^2+4)}\,dx$,

(d) P.V. $\displaystyle\int_{-\infty}^{\infty} \frac{2x+3}{x^2+2x+2}\,dx$.

Laurent series

Problem 42 Find the Laurent expansion of the function $f(z) = \dfrac{1}{z^2-3z+2}$ in each of the following regions:

(a) $\{z \in \mathbb{C} : |z| < 1\}$,

(b) $\{z \in \mathbb{C} : 1 < |z| < 2\}$,

(c) $\{z \in \mathbb{C} : |z| > 2\}$,

(d) $\{z \in \mathbb{C} : 0 < |z-1| < 1\}$,

(e) $\{z \in \mathbb{C} : |z-1| > 1\}$.

Problem 43 Expand each of the functions in a Laurent series about the indicated point z_0. Use the Laurent series to calculate $\mathrm{Res}(f, z_0)$.

(a) $f(z) = \dfrac{\sin z}{z^4}$, $z_0 = 0$,

(b) $f(z) = \dfrac{e^z}{z^2-1}$, $z_0 = 1$,

(c) $f(z) = \sin\left(\dfrac{1}{z}\right)$, $z_0 = 0$.